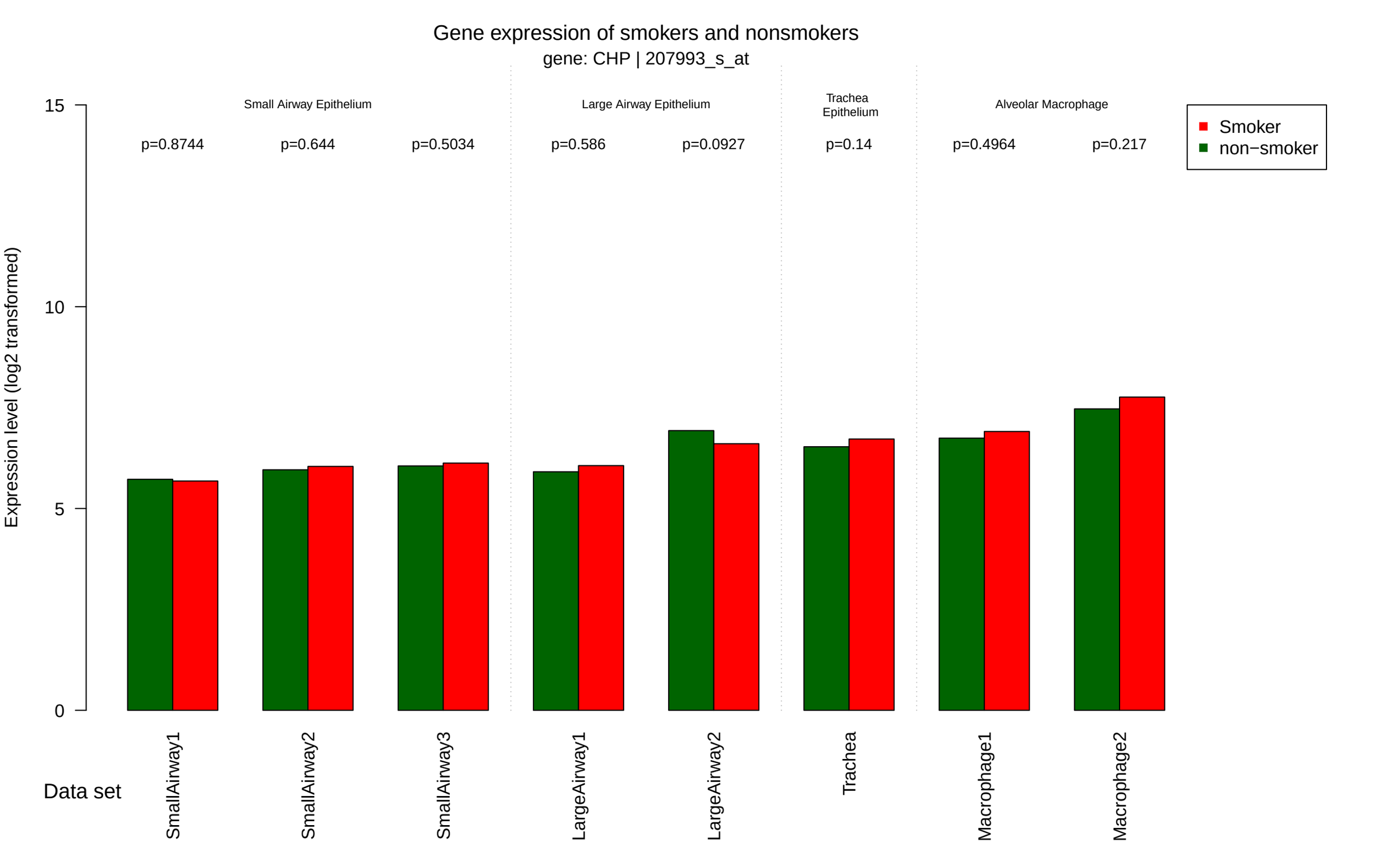
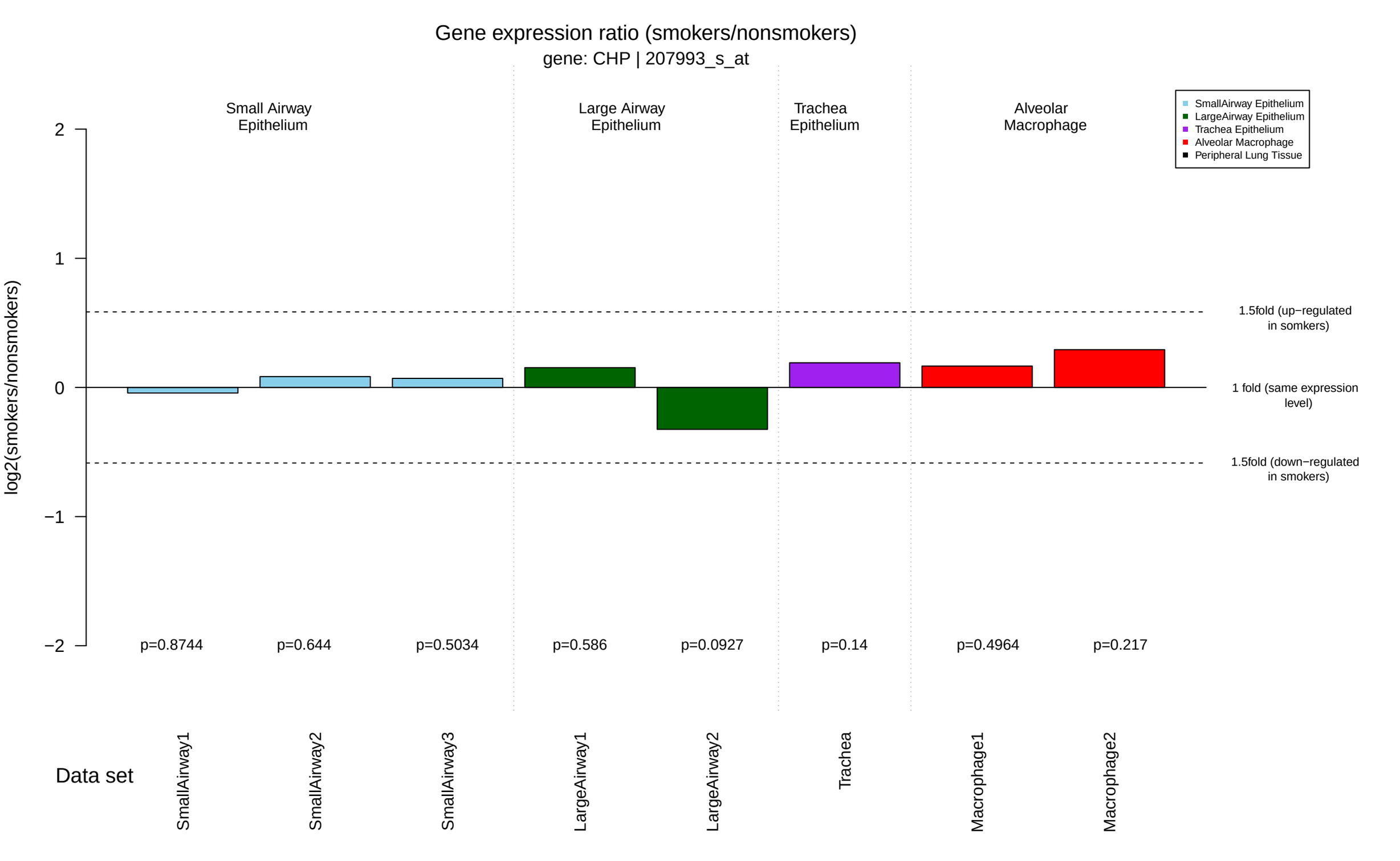
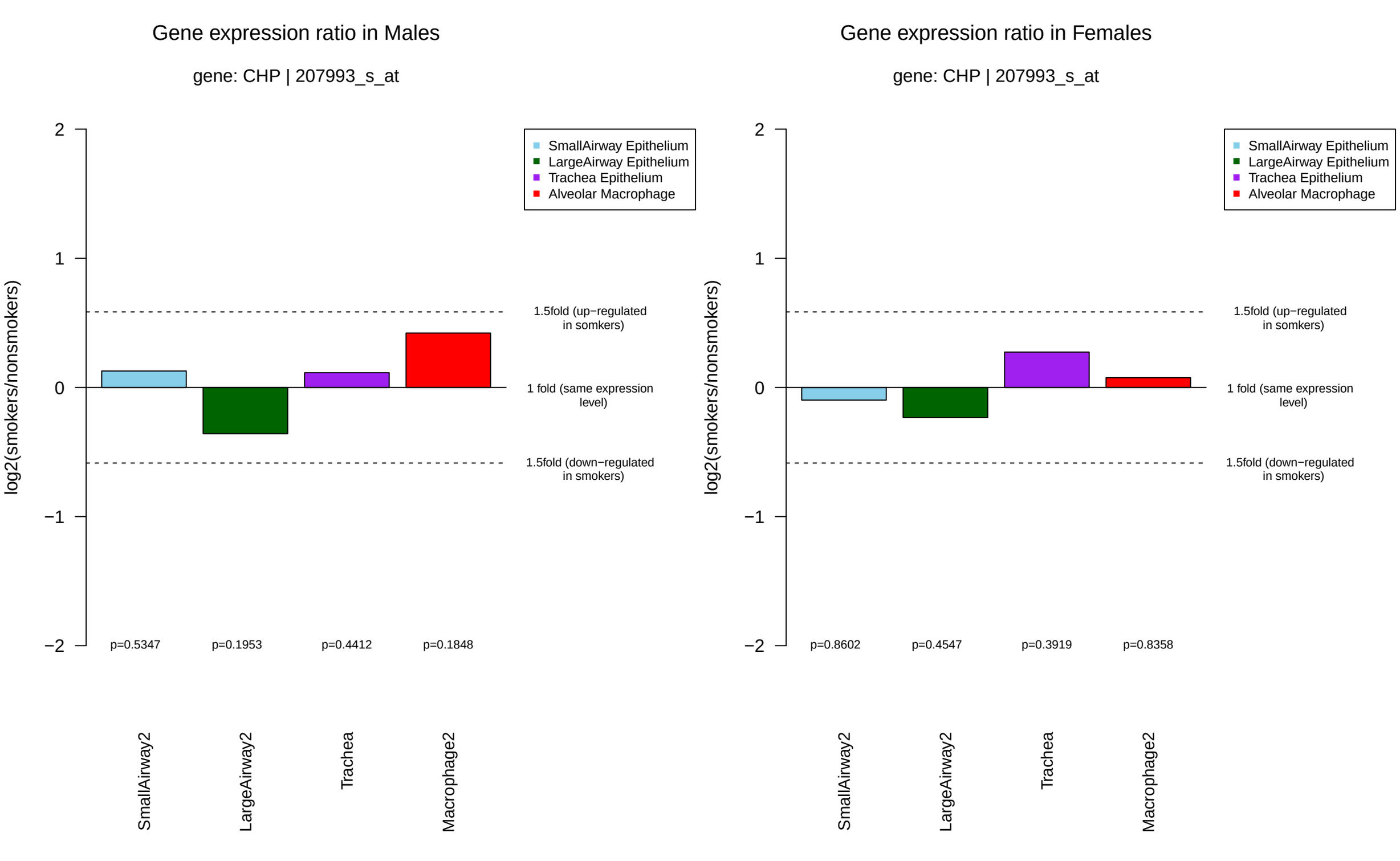
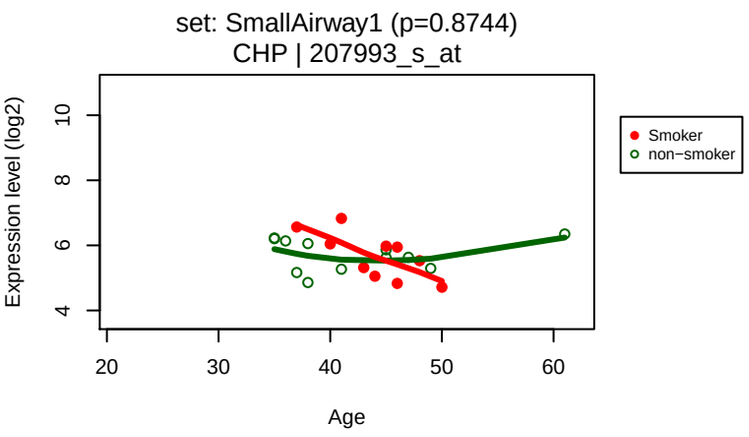
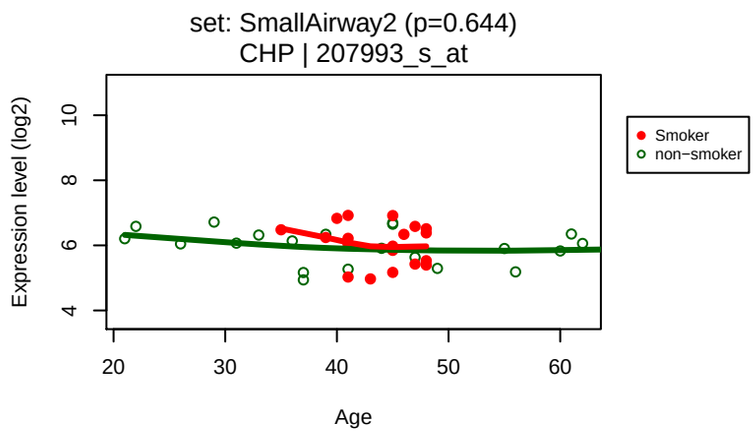
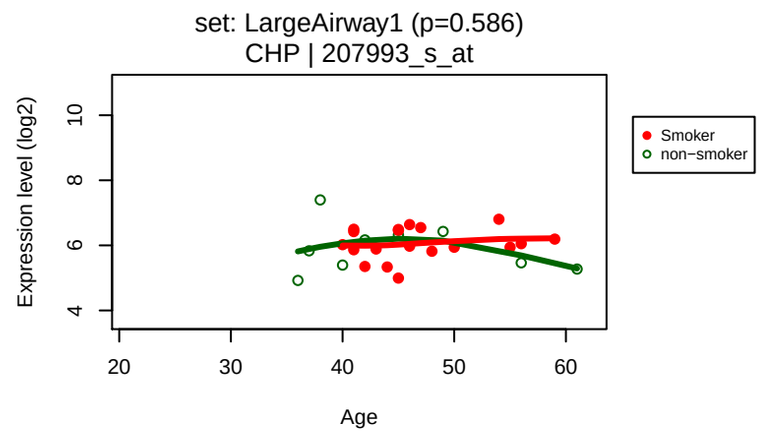

Gene expression of smokers and nonsmokers
gene: CHP | 207993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8744
p=0.644
p=0.5034
p=0.586
p=0.0927
p=0.14
p=0.4964
p=0.217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

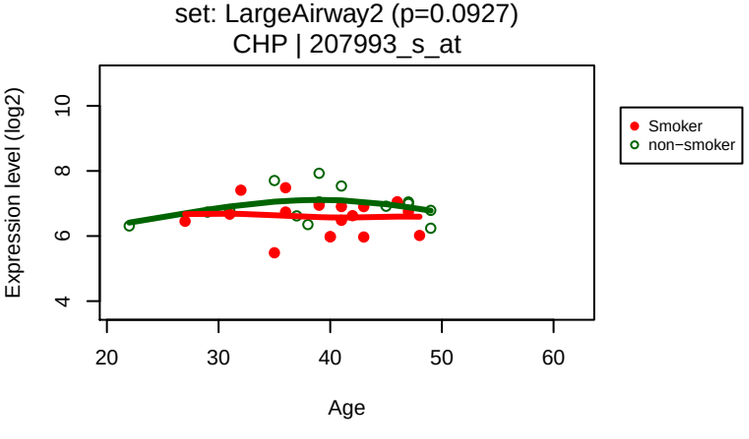

Gene expression ratio (smokers/nonsmokers)
gene: CHP | 207993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8744
p=0.644
p=0.5034
p=0.586
p=0.0927
p=0.14
p=0.4964
p=0.217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

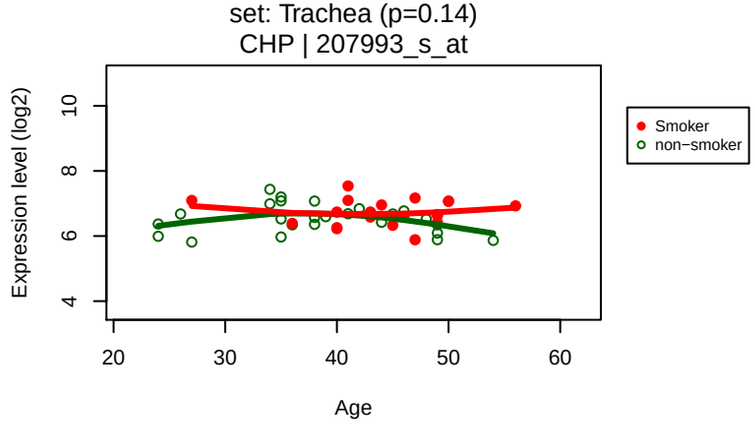

Gene expression ratio in Males
gene: CHP | 207993_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5347
p=0.1953
p=0.4412
p=0.1848
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHP | 207993_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8602
p=0.4547
p=0.3919
p=0.8358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

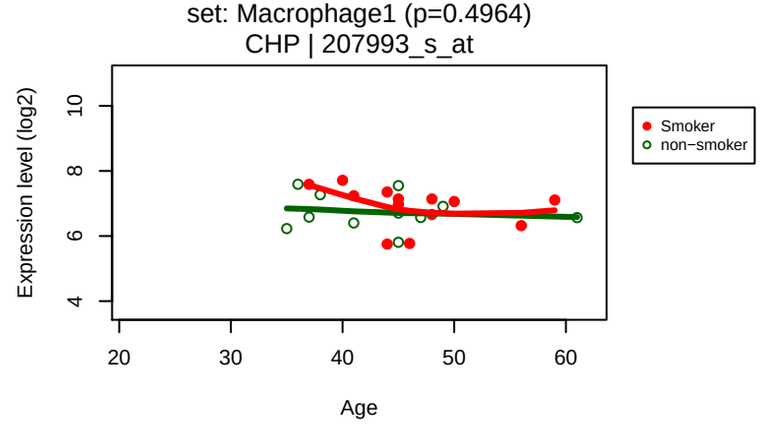

set: SmallAirway1 (p=0.8744)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker

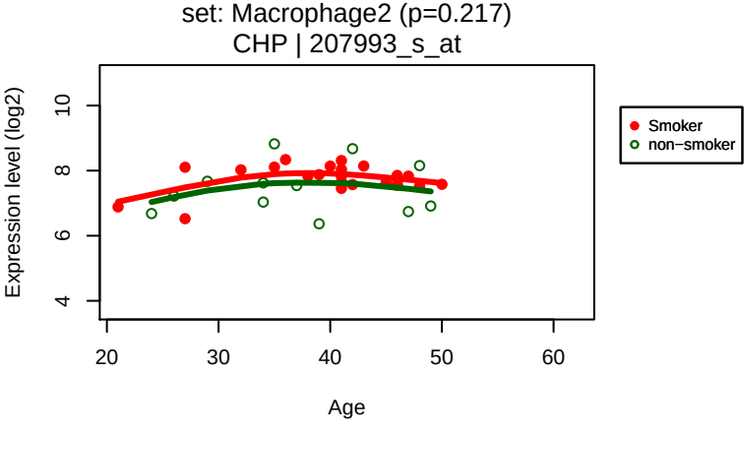

set: SmallAirway2 (p=0.644)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker

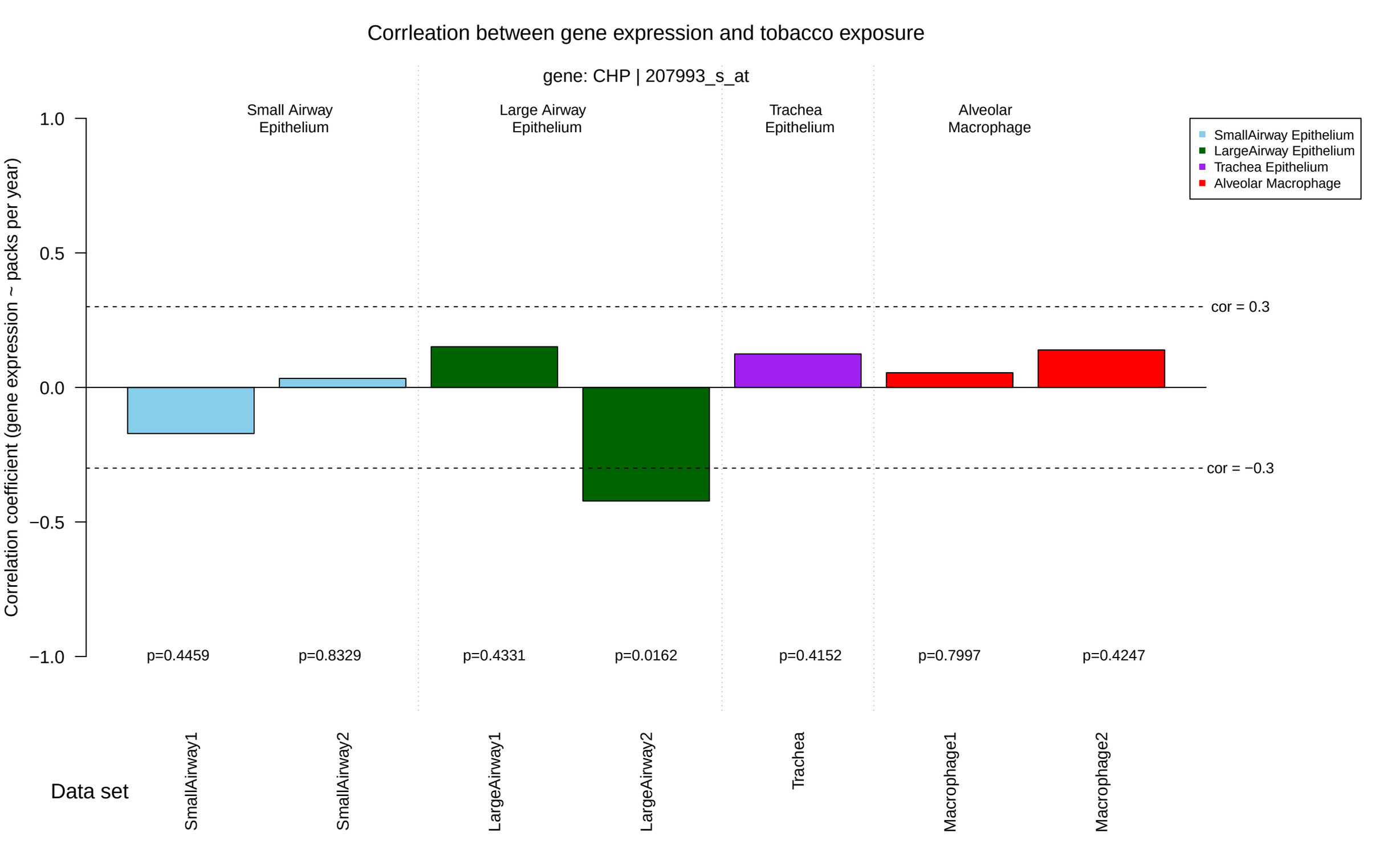

set: LargeAirway1 (p=0.586)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0927)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.14)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4964)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.217)
CHP | 207993_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHP | 207993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4459
p=0.8329
p=0.4331
p=0.0162
p=0.4152
p=0.7997
p=0.4247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2